

EXPERIMENT - 9

Title

Develop an AngularJS Form to Apply CSS and Event Handling

Aim

To develop an **AngularJS** application that:

- Creates a form
 - Applies dynamic CSS styles
 - Handles user events using AngularJS directives
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Prerequisites

- Basic knowledge of HTML, CSS, JavaScript
 - Understanding of AngularJS basics
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Software Requirements

- Operating System: Windows / Linux / macOS
 - Browser: Google Chrome
 - Code Editor: VS Code
 - AngularJS CDN
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Hardware Requirements

- Processor: Intel i3 or above
- RAM: Minimum 4 GB

Theory

What is AngularJS?

AngularJS is a JavaScript framework developed by Google for building dynamic web applications.

Key Features:

- Two-way data binding
- MVC architecture
- Directives (ng-model, ng-click, ng-style, ng-class)
- Event handling

CSS in AngularJS

AngularJS allows dynamic CSS using:

- ng-style
- ng-class

Event Handling in AngularJS

Events are handled using:

- ng-click
 - ng-change
 - ng-mouseover
 - ng-blur
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Algorithm

1. Create an HTML file.
 2. Include AngularJS CDN.
 3. Define AngularJS module and controller.
 4. Create a form with input fields.
 5. Apply dynamic CSS using AngularJS directives.
 6. Handle button click event.
 7. Display output dynamically.
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Project Structure

fsd_exp9/
|— index.html

Source Code

index.html

```
<!DOCTYPE html>
<html ng-app="myApp">
<head>
  <title>AngularJS Form with CSS & Events</title>

  <!-- AngularJS CDN -->
  <script
src="https://ajax.googleapis.com/ajax/libs/angularjs/1.8.2/angular.min.js"></s
cript>

  <style>
    .success {
      color: white;
      background-color: green;
      padding: 10px;
    }

    .error {
      color: white;
      background-color: red;
      padding: 10px;
```

```

    }

    input {
        padding: 5px;
        margin: 5px;
    }
</style>
</head>

<body ng-controller="formController">

    <h2>Registration Form</h2>

    <form>
        Name:
        <input type="text" ng-model="name" ng-change="checkName()" required />
        <br>

        Email:
        <input type="email" ng-model="email" required />
        <br><br>

        <button type="button" ng-click="submitForm()">Submit</button>
    </form>

    <br>

    <div ng-class="messageClass" ng-show="message">
        {{message}}
    </div>

<script>
    var app = angular.module("myApp", []);

    app.controller("formController", function($scope) {

        $scope.message = "";

        $scope.checkName = function() {
            if ($scope.name && $scope.name.length < 3) {
                $scope.message = "Name must be at least 3 characters";
                $scope.messageClass = "error";
            } else {
                $scope.message = "";
            }
        };

        $scope.submitForm = function() {
            if ($scope.name && $scope.email) {
                $scope.message = "Form Submitted Successfully!";
                $scope.messageClass = "success";
            } else {
                $scope.message = "Please fill all fields!";
                $scope.messageClass = "error";
            }
        };
    });

```

```
});  
</script>  
  
</body>  
</html>
```

Execution Steps

1. Save file as index.html
2. Open file in browser
3. Enter name and email
4. Observe:
 - CSS applied dynamically
 - Events triggered on input change
 - Button click event handled

Functionality Demonstrated

Feature	AngularJS Directive Used
Two-way Data Binding	ng-model
Event Handling	ng-click, ng-change
Dynamic CSS	ng-class
Conditional Display	ng-show

Sample Output

- ✓ If name < 3 characters → Red error message
 - ✓ If form valid → Green success message
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Expected Result

The AngularJS form successfully demonstrates:

- Dynamic CSS styling
 - Event handling
 - Two-way data binding
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Result

Thus, the AngularJS application was successfully developed to apply CSS styles dynamically and handle user events.

Viva Voce Questions

1. What is AngularJS?
 2. What is two-way data binding?
 3. What is ng-model?
 4. Difference between ng-style and ng-class?
 5. What is ng-click?
 6. What is a controller in AngularJS?
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Conclusion

This experiment demonstrates how AngularJS simplifies dynamic UI updates using data binding, event handling, and CSS manipulation.